Exploring Knowledge Distillation with Language-Specific Calibration Data

Alan Ispani and **Casper Anton Poulsen** and **Johan Laursen** and **Raghav Vacher**
IT University of Copenhagen
{alai,caap,jocl,ragv}@itu.dk

Abstract

In recent years, the development of large language models has rapidly progressed, achieving remarkable capabilities. A significant trend in this field is the effort to reduce the size of these models while maintaining their effectiveness. However, most advancements have predominantly centered around the English language. This paper shifts the focus to the distillation of the BLOOM and LLaMA language models using Spanish-only, as well as Spanish/English datasets by using the MiniLLM framework. Our objective is to evaluate the performance of these distilled models in comparison with an English-only distilled counterpart. The study aims to ascertain whether the MiniLLM algorithm successfully distills the larger model, or if the reduced parameters stem from the smaller model surrendering multilingual parameters to English.

1 Introduction

Large Language Models (LLMs) have proved to be a groundbreaking paradigm in Natural Language Processing. The top positions of NLP benchmarks like GLUE (Wang et al., 2018) and SuperGLUE (Wang et al., 2020a) are dominated by models with parameters counts in the billions (Zhong et al., 2022). Services like ChatGPT have also become engraved into the public consciousness, setting the record for fastest-growing user base (Hu, 2023). This success has driven researchers and businesses to adopt these technologies to remain competitive. However, closed-source models like GPT-4 (OpenAI, 2023) and Gemini (Team, 2023) present challenges such as high costs, lack of transparency, and data privacy concerns, making them less ideal for many academic and enterprise applications.

To address these challenges, many are turning to open-source or white-box LLMs like LLaMA (Touvron et al., 2023) or BLOOM (Workshop et al., 2022). Hosting these models resolves data privacy issues but still requires significant resources for self-hosting and fine-tuning. Furthermore, in many instances, a model capable of performing a wide array of NLP tasks isn't necessary; a more specialized model is often more desirable.

To achieve this specialization with a reduced parameter count, Knowledge Distillation is a viable method. Our paper explores this approach using the Microsoft MiniLLM framework (Gu et al., 2023a) and applying it to distill both the LLaMA and BLOOM models. Specifically, we focus on using the OASST1 instruction-following dataset (Köpf et al., 2023) to specialize these models in processing instructions Spanish and English. We also extend MiniLLM to support BLOOM and share our code for this implementation at https://github.com/CasperAntonPoulsen/KnowledgeDistillation

This research aligns with our broader objective of addressing the challenges posed by the size and complexity of models like GPT-4 and Gemini, especially in terms of accessibility, computational cost, and adaptability for specific linguistic contexts. By focusing on distilling large, multilingual models into more compact, language-specific versions, our study investigates whether a distilled model specialized in Spanish (or a combination of Spanish and English) can outperform an equivalent model distilled solely for English.

This exploration leads us to our central research question: "How much can we improve a language model's performance on a single language by distilling a larger language model using calibration data from that language?" Addressing this question is crucial, given the dominance of English in LLM development. Our study seeks to determine whether dedicated, language-specific distillation can produce models that are not only more compact but also more efficient in their targeted linguistic domain. Through this, we contribute to the ongoing discussion about the adaptability and efficiency of LLMs, especially in multilingual settings.

2 Related Work

The concept of knowledge distillation (KD) has evolved significantly since its initial proposition. The foundational work by Buciluǎ et al. (2006) introduced the idea of compressing the knowledge from an ensemble of models into a single model. This was achieved by sampling from a non-parametric estimate of the ensemble's joint distribution. Building upon this, Yu et al. (2013) and Hinton et al. (2015) refined the approach by shifting the focus towards minimizing the Kullback-Leibler Divergence (KLD) (Kullback and Leibler, 1951), a method of measuring how one probability distribution diverges from a second, expected probability distribution.

Hinton et al. (2015) further coined the term *Knowledge Distillation*, which has since become a cornerstone in numerous Natural Language Processing applications, particularly in text classification. This term encapsulates the process of training a smaller, more efficient 'student' model to mimic the performance of a larger, more complex 'teacher' model. Following this trajectory, recent works such as Song et al. (2020) and Zhang et al. (2023) have focused on training a student model to emulate the teacher's output distribution difference. Other researchers, like Sun et al. (2019) and Jiao et al. (2020), have explored mimicking the hidden states of the teacher model, while Wang et al. (2020b) have concentrated on replicating the attention scores.

For text generation, the prevailing approach for KD is to approximately minimize the forward KLD between the student model and the teacher model's generation distribution. This is typically done by either using the teacher model's output at each timestep as a supervisory signal, as demonstrated in the work of Sanh et al. (2019), or through direct training on the teacher model's generations, as explored by Kim and Rush (2016), Taori et al. (2023), and Chiang et al. (2023). However, this method is not without its limitations. As highlighted by Huszár (2015), minimizing forward KLD can lead to what is known as 'zero-forcing' behavior. In this scenario, the models attempt to encompass all modes of the target distribution, often at the expense of accurately capturing the more significant, prevalent modes.

To address this issue, Jiang et al. (2020) proposed the use of reverse KLD, which offers a different perspective by focusing on the student model's ability to learn the most critical aspects of the teacher's distribution. Building upon this idea, Gu et al. (2023b) adapted the reverse KLD approach specifically for the KD of large language models. This adoption of reverse KLD represents a notable innovation in KD techniques for LLMs. It prioritizes the accurate replication of essential characteristics in language models, thereby addressing the challenges of precision and model efficiency in complex text generation tasks.

Mukherjee and Hassan Awadallah (2020) perform KD for massive multilingual Named Entity Recognition with mBERT as the teacher and either a BiLSTM or Transformer Encoder as the student. They achieve 35x compression and 51x speedup over mBERT while retaining nearly 95% of its performance. To our knowledge, no explicit multilingual metrics have been calculated for knowledge distillation of large generative language models.

3 Data

Gu et al. (2023a) utilise two pre-training data sets in their MiniLLM pipeline, namely OpenWebText (Gokaslan and Cohen, 2019) for GPT-2 family models and the RoBERTa training corpus (Liu et al., 2019) for other models, which are then fine-tuned on the `databricks-dolly-15k` dataset (Conover et al., 2023).

In this work, the corpora utilised for pre-training are the Spanish and English Wikipedia datasets from huggingface[1], specifically the September 2023 versions. They consist of 1.52 and 1.55 million articles respectively. The model is then calibrated on the OpenAssistant Conversations Dataset (OASST1) (Köpf et al., 2023), which is a human-generated, human-annotated assistant-style conversation corpus annotated with quality ratings and over 10,000 fully annotated conversation trees in 35 different languages. The corpus is a product of a worldwide crowd-sourcing effort involving over 13,500 volunteers, consisting of 160,000 messages in 35 different languages annotated with 460,000 quality ratings.

The OASST1 dataset was chosen due to its composition of human-generated Spanish and English language content, which closely mirrors the instruction-based format and content characteristics of the `databricks-dolly-15k` dataset.

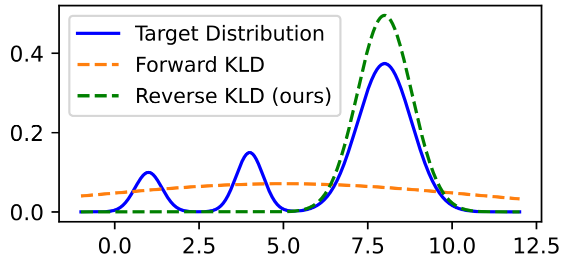


Figure 1: Toy experiment where a Gaussian mixture is fitted with a Gaussian Distribution using forward and reverse KLD. Figure taken from Gu et al. (2023b)

4 Methods

In this study, we focus on **Knowledge Distillation (KD)**, a technique aimed at transferring knowledge from a large, computationally expensive "teacher" model to a smaller, more efficient "student" model. For conditional language generation tasks, where a model generates a response $\boldsymbol{y}$ based on a given prompt $\boldsymbol{x}$, sampled from a distribution $p_{\boldsymbol{x}}$, the objective is to minimize the discrepancy between the probability distributions of the teacher and student models, denoted as $p(\boldsymbol{y}|\boldsymbol{x})$ and $q_\theta(\boldsymbol{y}|\boldsymbol{x})$ respectively, where θ represents the parameters of the student model.

Traditionally, this discrepancy is quantified using the *forward* Kullback-Leibler Divergence (KLD), formulated as:

$$KL[p||q_\theta] = \sum_{x \in X} p(y|x) \log \left(\frac{p(y|x)}{q_\theta(y|x)} \right) \quad (1)$$

However, in language generation tasks, forward KLD tends to overestimate the sparsely populated regions of the teacher model's probability distribution p, particularly when the student model's distribution q_θ lacks the expressiveness to capture all the modes of p. This issue is highlighted by Huszár (2015). To address this, the **MiniLLM** algorithm (Gu et al., 2023b) adopts the *reverse* KLD, defined as:

$$KL[q_\theta||p] = \sum_{x \in X} q_\theta(y|x) \log \left(\frac{q_\theta(y|x)}{p(y|x)} \right) \quad (2)$$

They argue that this leads to a mode-seeking behaviour, where q_θ assigns high probabilities to p's large modes, ignoring the small ones (see Figure 1 for a toy experiment that illustrates the difference).

Additionally, they incorporate a single-step regularization strategy, as recommended by Czarnecki et al. (2019). This strategy is critical in maintaining the generation quality of individual steps in the model. It effectively mitigates the accumulation of errors in the initial tokens across a sentence, thereby reducing training variance and accelerating the model's convergence. They combine this with length normalization to discourage the model from producing very short responses.

To avoid reward-hacking, a phenomenon where the student model occasionally produces degenerate sentences $\boldsymbol{y}$ (e.g. repeated phrases) that are erroneously scored highly by the teacher model, MiniLLM utilizes teacher-mixed sampling. This technique sums the teacher and student distributions at each time step and helps alleviate reward-hacking by suppressing low-quality generation.

In the final phase of the training process, language modelling loss is applied to the parameter update by computing the gradient of the pre-training corpus' loss during gradient descent. This helps the student model preserve its language modelling abilities while aligning more closely to the teacher model in instruction following tasks.

For a full breakdown of the MiniLLM algorithm, please see section 2.3 of Gu et al. (2023b).

Since MiniLLM is extremely resource intensive, we apply three changes that allow us to adapt their methodology to our available hardware. Firstly, we reduce the sequence length from 512 to 256 tokens. This potentially reduces the model's performance at longer sequences. Secondly, we reduce the amount of training steps from 5000 to 1000, as our initial exploration showed that models don't improve significantly after that point, and Gu et al. (2023b) show that the KLD between the teacher and student models plateaus at that point (see Figure 6 in their paper). Thirdly, we utilize ZeRO-Offload (Ren et al., 2021). This framework allows optimizer and parameter offloading to CPU, and is designed to minimize data movement to/from GPU, allowing training of large models on a single GPU.

5 Experiments/Analysis

5.1 Languages for Evaluation

Our objective was to assess how distillation impacts the performance of the model when specialized in different languages. We specifically wanted to see how performance on English and Spanish changed when either or both languages were used in the

[1] https://huggingface.co/datasets/graelo/wikipedia

	BLOOM												
	headqa		lambada		paws			xnli			xstorycloze		
	en	es	en	es	en	es	zh	en	es	zn	en	es	zh
7b	31.2	29.5	57.6	36.6	40.1	41.7	52.0	53.6	48.4	35.2	70.6	66.1	61.9
3b	28.4	26.5	51.8	34.6	**43.8**	42.1	52.7	53.1	47.8	37.4	**66.8**	**64.0**	**61.0**
3b<7b english	28.0	26.5	51.3	**35.0**	41.9	44.3	54.4	54.2	**49.8**	38.5	66.4	62.3	58.6
3b<7b spanish	**28.5**	26.5	**52.4**	34.2	39.7	47.1	54.1	52.7	49.6	**40.0**	66.2	62.7	58.2
3b<7b both	27.9	**26.9**	50.2	31.1	40.3	**48.1**	**54.9**	**54.7**	49.1	39.2	66.0	63.2	58.9
	LLaMA												
	headqa		lambada		paws			xnli			xstorycloze		
	en	es	en	es	en	es	zh	en	es	zh	en	es	zh
13b	37.3	32.4	71.6	43.7	47.2	45.5	54.3	35.7	33.5	34.3	79.2	71.4	59.2
7b	**36.5**	**30.7**	70.7	40.3	**38.1**	42.3	**49.5**	51.6	44.7	35.5	**79.9**	**69.0**	**56.7**
7b<13b english	35.6	29.7	**74.7**	**45.2**	37.1	40.1	48.4	**56.8**	**44.9**	**35.8**	78.2	66.6	55.7
7b<13b spanish	35.2	29.3	73.6	42.1	35.9	41.3	48.6	50.0	41.9	35.4	77.8	66.9	54.9
7b<13b both	35.4	29.5	74.4	39.8	35.6	**44.8**	48.5	55.9	39.8	34.0	78.6	67.2	55.3

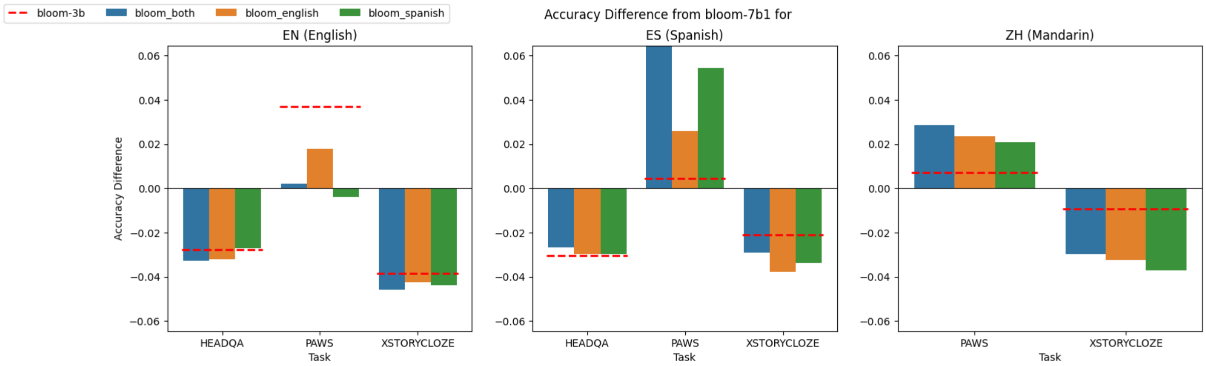
Table 1: Accuracy metrics for the base and distilled models. Scores in bold are the highest for the given tasks

knowledge distillation. Chinese evaluation tasks were also chosen when available to quantify the distillation effect on a different language family.

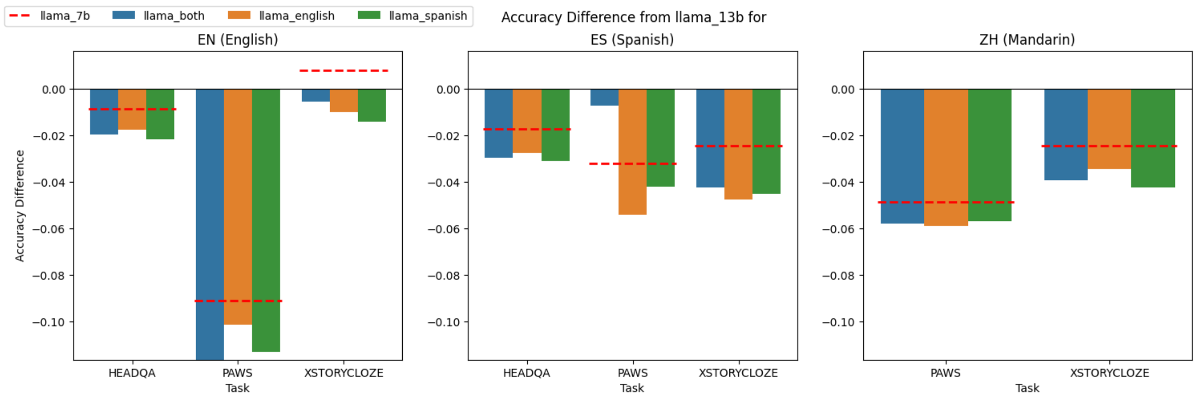
5.1.1 Evaluation: Zero-Shot framework

All of our zero-shot tasks were run using the Language Model Evaluation Harness Library (Gao et al., 2021). The library offers a wide range of tasks and several of them have multilingual options. The framework provides an easy to use command line interface for testing pretrained models with over 200 evaluation tasks. We chose 5 of the 6 available tasks that had at least both English and Spanish versions. Of the five benchmarks, only HeadQA and LAMBADA do not have a Chinese variant. The only multilingual task we didn't include was the Multilingual Grade School Math (MGSM) (Shi et al., 2022), as we believed it was too domain specific to math knowledge. An example of each task is shown in Table 2 in the Appendix, with an explanation of how lm-eval performs the evaluation. The code used to evaluate is in the following github repo: https://github.com/johanlaursen/lmeval.

- **PAWS-X** (Yang et al., 2019) is a human translated version of the PAWS (Zhang et al., 2019) paraphrase identification task with 23,659 human translated pairs available in six languages.
- **XStoryCloze** (Lin et al., 2022) is a professionally translated version of the English StoryCloze (Mostafazadeh et al., 2016) dataset. The task is a story understanding task where the language model is given a four sentence story and must choose the correct story conclusion from two options. The task is a form of textual entailment where the correct choice can be seen as the entailment and the wrong choice as the contradiction.
- **XNLI** (Conneau et al., 2018) is a textual entailment task where the model has to determine if the hypothesis sentence is an "entailment", "neutral" or "contradiction" of the premise.
- **HeadQA** (Vilares and Gómez-Rodríguez, 2019) is a multi-choice question answering dataset. The questions come from Spanish healthcare exams. This task is especially challenging even for expert humans due to the extensive knowledge required.
- **LAMBADA** (Paperno et al., 2016) is a word prediction task where the metric is perplexity. The dataset is designed so that the target word is difficult to predict when only the latest sentence is available but easy to guess when broader context is given.



(a) BLOOM Model Results



(b) LLaMa Model Results

Figure 2: Accuracy differences in distilled models with teacher models across English, Spanish, and Mandarin. The red dashed line denotes the baseline accuracy of the original 7 billion parameter LLaMa and 3 billion parameter BLOOM models. If our hypothesis was true, we would for example see an increase in performance for the Spanish-distilled model in Spanish tasks. The patterns in the figure don't indicate that Knowledge Distillation improves a smaller model's natural language ability when calibrated on instruction-following data. Note: HeadQA performance is excluded for Mandarin as it contains no evaluation for it. LAMBADA and XNLI are excluded from the visualizations as neither of them is designed as an instruction-following task.

6 Results and Discussion

Table 1 shows the accuracy of the original and distilled models on our selected evaluation benchmarks. It illustrates the performance of different models across English (en), Spanish (es) and Mandarin (zh). We compare these distilled models to the original 3 billion parameter BLOOM and 7 billion parameter LLaMA models to ascertain the degree to which multilinguality and language specific abilities are retained in the model based on the calibration language used.

Looking at Table 1 and Figures 2a and 2b, we can see that the distillation process tends to reduce or maintain performance of the original model in the chosen NLP tasks. Some notable exceptions are:

- Mandarin and Spanish PAWS tasks for BLOOM, where even though the student model is already better than the teacher the distillation seems to improve its performance regardless of which calibration data was used.

- Spanish PAWS task for LLaMA, where only the model distilled on a combination of English and Spanish calibration data gets improved performance.

If language-specific calibration improved performance on tasks of that language, we would most likely see an increase in accuracy for the model distilled on that specific language compared to the original. From these results, language calibration data and performance on that language's tasks don't seem to be correlated. Therefore, our results do not support the theory that Knowledge Distillation improves the smaller model's natural language ability when calibrated on instruction-following data.

7 Limitations

Our research employs a specific finetuning dataset designed to enhance the model's ability to follow instructions. This choice, however, presents challenges in accurately evaluating the model's performance using our chosen NLP metrics. Notably, these metrics do not seem to reflect improvements from our dataset, suggesting a potential mismatch between the finetuning data and the evaluation criteria. A more targeted finetuning approach, aligned with these metrics, might yield more relevant results.

Furthermore, our findings differ from those in Gu et al. (2023b), possibly because our benchmarks, do not exclusively focus on instruction following. The model's learned output distribution, reshaped by the finetuning process, could affect its performance on these tasks. For instance, it might assign lower probabilities to tokens not resembling typical instruction responses. Moreover, verifying the correctness of a sentence is inherently easier than generating a correct sentence (Lightman et al., 2023). Our evaluation methodology currently falls short in this aspect, as it focuses solely on assessing the model's perceived likelihood of a sentence being correct, without challenging the model to generate outputs, which is a more complex task.

8 Future Work

Given the methodological concerns raised in Section 7, a better way of testing the knowledge distillation process would be to use an evaluation that tests for instruction-following generation accuracy, such as Super-Natural Instructions (Wang et al., 2022), which is a benchmark comprising 1616 diverse NLP tasks with expert-written instructions designed to train and evaluate models on following in-context instructions. This dataset would allow us to measure the difference in generation performance between the models distilled with Spanish vs. English calibration data.

Each task is evaluated in two-shot, and consists of a prompt with:

- A problem definition, such as "Given a paragraph, your job is to generate a question that can be answered from the passage. The answer to your question should be a single entity, person, time, etc. that can be extracted from the passage".
- Two positive examples, where an example input and correct output are given.

We could choose Spanish to Spanish, English to English and Chinese to Chinese tasks, and use the same metric as Gu et al. (2023b) to measure the precision of the model generation, namely the Rouge-L score. This would make it easy to compare our results to theirs. Wang et al. (2022) have also shown that Rouge-L is suitable for large-scale instruction-following evaluation.

Alternatively, in order to properly use our chosen evaluation metrics, a comparison between direct finetuning and knowledge distillation could be done to demonstrate the effectiveness of using a teacher model during the finetuning process. However, a fair experimental setup with this goal in mind is not easy to achieve as the knowledge distillation process requires significant overhead. Simply limiting the number of iterations or epochs would not take into account how much more computationally expensive it is to run the MiniLLM algorithm, which requires running inference on a larger teacher model, as well as calculating the reverse KLD between the models' output distributions. The end result of this comparison would be a better understanding of the effectiveness of using knowledge distillation, instead of finetuning, to improve a model's performance on a target language.

9 Conclusion

For this project we extended MiniLLM by adding BLOOM to its roster of models. We used this along with MiniLLMs LLaMA implementation to perform Knowledge Distillation. We explored whether using Spanish-specific calibration data would improve the distilled model's performance on a variety of NLP benchmarks.

Our findings indicate a lack of empirical evidence to support the hypothesis that Knowledge Distillation enhances the linguistic capabilities of smaller models, particularly when these models are calibrated with instruction-following data. For instance, the performance of models distilled with Spanish or a combination of English and Spanish calibration data did not consistently outperform those distilled solely with English data in Spanish language tasks.

We have laid out multiple paths in this study that we strongly believe could find a more sound empirical basis to support our original hypothesis.

References

Cristian Buciluǎ, Rich Caruana, and Alexandru Niculescu-Mizil. 2006. Model compression. In *Proceedings of the 12th ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 535–541.

Wei-Lin Chiang, Zhuohan Li, Zi Lin, Ying Sheng, Zhanghao Wu, Hao Zhang, Lianmin Zheng, Siyuan Zhuang, Yonghao Zhuang, Joseph E Gonzalez, et al. 2023. Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality. *See https://vicuna. lmsys. org (accessed 14 April 2023)*.

Alexis Conneau, Ruty Rinott, Guillaume Lample, Adina Williams, Samuel Bowman, Holger Schwenk, and Veselin Stoyanov. 2018. XNLI: Evaluating cross-lingual sentence representations. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2475–2485, Brussels, Belgium. Association for Computational Linguistics.

Mike Conover, Matt Hayes, Ankit Mathur, Jianwei Xie, Jun Wan, Sam Shah, Ali Ghodsi, Patrick Wendell, Matei Zaharia, and Reynold Xin. 2023. Free dolly: Introducing the world's first truly open instruction-tuned llm.

Wojciech Marian Czarnecki, Razvan Pascanu, Simon Osindero, Siddhant M. Jayakumar, Grzegorz Swirszcz, and Max Jaderberg. 2019. Distilling policy distillation.

Leo Gao, Jonathan Tow, Stella Biderman, Sid Black, Anthony DiPofi, Charles Foster, Laurence Golding, Jeffrey Hsu, Kyle McDonell, Niklas Muennighoff, Jason Phang, Laria Reynolds, Eric Tang, Anish Thite, Ben Wang, Kevin Wang, and Andy Zou. 2021. A framework for few-shot language model evaluation.

Aaron Gokaslan and Vanya Cohen. 2019. Openwebtext corpus. http://Skylion007.github.io/OpenWebTextCorpus.

Yuxian Gu, Li Dong, Furu Wei, and Minlie Huang. 2023a. Knowledge distillation of large language models. *arXiv preprint arXiv:2306.08543*.

Yuxian Gu, Li Dong, Furu Wei, and Minlie Huang. 2023b. Knowledge distillation of large language models. *arXiv preprint arXiv:2306.08543*.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the knowledge in a neural network.

Krystal Hu. 2023. Chatgpt sets record for fastest-growing user base - analyst note.

Ferenc Huszár. 2015. How (not) to train your generative model: Scheduled sampling, likelihood, adversary?

Haoming Jiang, Pengcheng He, Weizhu Chen, Xiaodong Liu, Jianfeng Gao, and Tuo Zhao. 2020. SMART: Robust and efficient fine-tuning for pre-trained natural language models through principled regularized optimization. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 2177–2190, Online. Association for Computational Linguistics.

Xiaoqi Jiao, Yichun Yin, Lifeng Shang, Xin Jiang, Xiao Chen, Linlin Li, Fang Wang, and Qun Liu. 2020. TinyBERT: Distilling BERT for natural language understanding. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 4163–4174, Online. Association for Computational Linguistics.

Yoon Kim and Alexander M Rush. 2016. Sequence-level knowledge distillation. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 1317–1327.

Solomon Kullback and Richard A Leibler. 1951. On information and sufficiency. *The annals of mathematical statistics*, 22(1):79–86.

Andreas Köpf, Yannic Kilcher, Dimitri von Rütte, Sotiris Anagnostidis, Zhi-Rui Tam, Keith Stevens, Abdullah Barhoum, Nguyen Minh Duc, Oliver Stanley, Richárd Nagyfi, Shahul ES, Sameer Suri, David Glushkov, Arnav Dantuluri, Andrew Maguire, Christoph Schuhmann, Huu Nguyen, and Alexander Mattick. 2023. Openassistant conversations – democratizing large language model alignment.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2023. Let's verify step by step.

Xi Victoria Lin, Todor Mihaylov, Mikel Artetxe, Tianlu Wang, Shuohui Chen, Daniel Simig, Myle Ott, Naman Goyal, Shruti Bhosale, Jingfei Du, Ramakanth Pasunuru, Sam Shleifer, Punit Singh Koura, Vishrav Chaudhary, Brian O'Horo, Jeff Wang, Luke Zettlemoyer, Zornitsa Kozareva, Mona Diab, Veselin Stoyanov, and Xian Li. 2022. Few-shot learning with multilingual generative language models. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 9019–9052, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. Roberta: A robustly optimized bert pretraining approach.

Nasrin Mostafazadeh, Nathanael Chambers, Xiaodong He, Devi Parikh, Dhruv Batra, Lucy Vanderwende, Pushmeet Kohli, and James Allen. 2016. A corpus and cloze evaluation for deeper understanding of commonsense stories. In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 839–849, San Diego, California. Association for Computational Linguistics.

Subhabrata Mukherjee and Ahmed Hassan Awadallah. 2020. XtremeDistil: Multi-stage distillation for massive multilingual models. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 2221–2234, Online. Association for Computational Linguistics.

OpenAI. 2023. Gpt-4 technical report.

Denis Paperno, Germán Kruszewski, Angeliki Lazaridou, Ngoc Quan Pham, Raffaella Bernardi, Sandro Pezzelle, Marco Baroni, Gemma Boleda, and Raquel Fernández. 2016. The LAMBADA dataset: Word prediction requiring a broad discourse context. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1525–1534, Berlin, Germany. Association for Computational Linguistics.

Jie Ren, Samyam Rajbhandari, Reza Yazdani Aminabadi, Olatunji Ruwase, Shuangyan Yang, Minjia Zhang, Dong Li, and Yuxiong He. 2021. Zero-offload: Democratizing billion-scale model training.

Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2019. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*.

Freda Shi, Mirac Suzgun, Markus Freitag, Xuezhi Wang, Suraj Srivats, Soroush Vosoughi, Hyung Won Chung, Yi Tay, Sebastian Ruder, Denny Zhou, Dipanjan Das, and Jason Wei. 2022. Language models are multilingual chain-of-thought reasoners.

Kaitao Song, Hao Sun, Xu Tan, Tao Qin, Jianfeng Lu, Hongzhi Liu, and Tie-Yan Liu. 2020. Lightpaff: A two-stage distillation framework for pre-training and fine-tuning.

Siqi Sun, Yu Cheng, Zhe Gan, and Jingjing Liu. 2019. Patient knowledge distillation for bert model compression. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 4323–4332.

Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B Hashimoto. 2023. Stanford alpaca: An instruction-following llama model.

Gemini Team. 2023. Gemini: A family of highly capable multimodal models.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. 2023. Llama: Open and efficient foundation language models.

David Vilares and Carlos Gómez-Rodríguez. 2019. HEAD-QA: A healthcare dataset for complex reasoning. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 960–966, Florence, Italy. Association for Computational Linguistics.

Alex Wang, Yada Pruksachatkun, Nikita Nangia, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R. Bowman. 2020a. Superglue: A stickier benchmark for general-purpose language understanding systems.

Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel Bowman. 2018. GLUE: A multi-task benchmark and analysis platform for natural language understanding. In *Proceedings of the 2018 EMNLP Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, pages 353–355, Brussels, Belgium. Association for Computational Linguistics.

Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. 2020b. Minilm: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. *Advances in Neural Information Processing Systems*, 33:5776–5788.

Yizhong Wang, Swaroop Mishra, Pegah Alipoormolabashi, Yeganeh Kordi, Amirreza Mirzaei, Atharva Naik, Arjun Ashok, Arut Selvan Dhanasekaran, Anjana Arunkumar, David Stap, Eshaan Pathak, Giannis Karamanolakis, Haizhi Lai, Ishan Purohit, Ishani Mondal, Jacob Anderson, Kirby Kuznia, Krima Doshi, Kuntal Kumar Pal, Maitreya Patel, Mehrad Moradshahi, Mihir Parmar, Mirali Purohit, Neeraj Varshney, Phani Rohitha Kaza, Pulkit Verma, Ravsehaj Singh Puri, Rushang Karia, Savan Doshi, Shailaja Keyur Sampat, Siddhartha Mishra, Sujan Reddy A, Sumanta Patro, Tanay Dixit, and Xudong Shen. 2022. Super-NaturalInstructions: Generalization via declarative instructions on 1600+ NLP tasks. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 5085–5109, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

BigScience Workshop, Teven Le Scao, Angela Fan, Christopher Akiki, Ellie Pavlick, Suzana Ilić, Daniel Hesslow, Roman Castagné, Alexandra Sasha Luccioni, François Yvon, et al. 2022. Bloom: A 176b-parameter open-access multilingual language model. *arXiv preprint arXiv:2211.05100*.

Yinfei Yang, Yuan Zhang, Chris Tar, and Jason Baldridge. 2019. PAWS-X: A cross-lingual adversarial dataset for paraphrase identification. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3687–3692, Hong Kong, China. Association for Computational Linguistics.

Dong Yu, Kaisheng Yao, Hang Su, Gang Li, and Frank Seide. 2013. Kl-divergence regularized deep neural network adaptation for improved large vocabulary speech recognition. In *2013 IEEE International Conference on Acoustics, Speech and Signal Processing*, pages 7893–7897. IEEE.

Rongzhi Zhang, Jiaming Shen, Tianqi Liu, Jialu Liu, Michael Bendersky, Marc Najork, and Chao Zhang. 2023. Do not blindly imitate the teacher: Using perturbed loss for knowledge distillation. *arXiv preprint arXiv:2305.05010*.

Yuan Zhang, Jason Baldridge, and Luheng He. 2019. Paws: Paraphrase adversaries from word scrambling.

Qihuang Zhong, Liang Ding, Yibing Zhan, Yu Qiao, Yonggang Wen, Li Shen, Juhua Liu, Baosheng Yu, Bo Du, Yixin Chen, Xinbo Gao, Chunyan Miao, Xiaoou Tang, and Dacheng Tao. 2022. Toward efficient language model pretraining and downstream adaptation via self-evolution: A case study on superglue.

A Task Examples

Benchmark	Example prompts	Objective
PAWS-X	The Tabaci River is a tributary of the River Leurda in Romania, right? Yes, The Leurda River is a tributary of the River Tabaci in Romania. The Tabaci River is a tributary of the River Leurda in Romania, right? No, The Leurda River is a tributary of the River Tabaci in Romania.	Loglikelihood rolling
XStoryCloze	Kathy went shopping. She found a pair of great shoes. The shoes were $300. She bought the shoes. Kathy hated buying shoes. Kathy went shopping. She found a pair of great shoes. The shoes were $300. She bought the shoes. She felt buyer's remorse after the purchase.	Loglikelihood
XNLI	But that takes too much planning, right? Yes, it doesn't take much planning. But that takes too much planning, right? Also, it doesn't take much planning. But that takes too much planning, right? No, it doesn't take much planning.	Loglikelihood rolling
HeadQA	Question: The pyramidal neurons are in which location? Answer: Cerebellum | Brain | Spinal cord | Spinal Ganglia	Loglikelihood
LAMBADA	Context: "Yes, I thought I was going to lose the baby." "I was scared too," he stated, sincerity flooding his eyes. "You were ?" "Yes, of course. Why do you even ask?" "This baby wasn't exactly planned for." "Do you honestly think that I would want you to have a ___? Answer: miscarriage	Loglikelihood

Table 2: Example English prompts for our five benchmarks. The differences between prompts for a single example are underlined. Loglikelihood objective is log probability of getting the underlined target text given the starting context. Loglikelihood rolling is the log probability of getting the entire sequence. LAMBADA calculates perplexity from the loglikelihood.

B Group Contributions

Alan worked on getting MiniLLM to work on the available hardware by troubleshooting various compatibility issues and applying ZeRO offload. This allowed LLaMA to be distilled with 7 times fewer VRAM than required in the original paper. He wrote the Related Work, Methods and Limitations sections and helped write the Introduction, Results/Discussion, Future Work and Conclusion sections.

Casper worked on extending MiniLLM to support BigScience's BLOOM model, as well as setting up and running the distillations of Llama and BLOOM. He also created the data cleaning and processing with **Alan** for the distillations. He also performed the evaluation of the distilled models using **Johans** implementation of LMEval. He also wrote the Introduction, results/discussion and helped with the conclusion and created the tables.

Johan worked on setting up LMEval and getting evaluations to work. Code contributions can be seen in https://github.com/johanlaursen/lmeval. Helped write discussion and future work and limitation sections as well. Wrote the Evaluation:Zero-shot framework section as well as the table of zero shot examples in the appendix.

Raghav generated visualizations for the final results across multiple languages from the evaluation done by **Casper**. Helped and contributed to writing the results, documenting the data sets utilized, providing relevant justifications for our choices, writing comprehensive captions, and writing the abstract.